A Experimental setup

partially observed clinical simulator · GRPO with alternative reward and training strategies · one common held-out verifier

Clinical multi-turn environment

refill triage · acute diagnostics · dosing titration

hidden state · diagnosis · severity · dose

MedGemma-4B-it
LoRA-adapted policy

ask

finding revealed

Patient simulator
hidden clinical state

commit

ask again · or commit

clinical action

◆ **Exact adaptive oracle**
reference procedure

◆ **Deterministic verifier**
terminal outcome scorer

policy rollouts

GRPO training

trained policies

reward or training strategy differs

Outcome-only
terminal reward only

Credit assignment
+ per-ask info credit

Decoupled
gather / commit split

Disaggregated RL

Trainer node
GRPO · LoRA updates

Rollout node
vLLM · 8 rollouts/prompt

policy weights → · ← rollout trajectories

Held-out evaluation

unseen simulator cases

Held-out simulator cases

Same pure verifier
one terminal score, every method

reward shaping off

metrics

overall value · high-risk value

question burden · oracle-grade support

B Same patient, different decision procedures

one held-out high-risk case (acute abdominal pain, hidden emergent severity) · same simulator, same final verifier · only the training method differs

single representative rollouts (k = 12 per condition), not modal or aggregate behaviour · oracle-grade reward does not establish that sufficient evidence was gathered

finding present · finding normal

Exact adaptive oracle
reference procedure · not a learned policy

Abdominal rigidity (present) → Rebound tenderness (present) → CT free air (present) → Lactate elevated (present) → **Emergency treatment**

+0.83 reward

4 targeted findings → emergency treatment

Outcome-only GRPO
1 finding · premature commitment

Abdominal rigidity (present) → **Admit**

+0.17 reward

emergency action missed

Credit-assignment GRPO
4 findings · successful sampled trajectory

Abdominal rigidity (present) → Guarding (present) → Rebound tenderness (present) → CT free air (present) → **Emergency treatment**

+0.83 reward

oracle-grade by reward; lactate, an emergent-defining finding, never requested

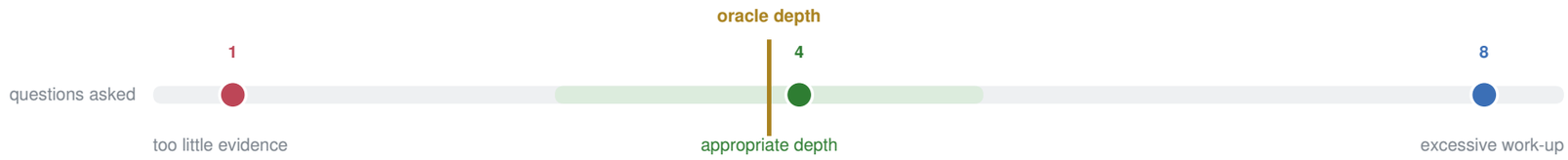
Decoupled explore/exploit
8 findings vs oracle 4 · correct final action

additional non-decisive findings

Abdominal rigidity (present) → Rebound tenderness (present) → Guarding (present) → Radiating pain → Severe dyspnea → Blood pressure → White cell count → ECG → **Emergency treatment**

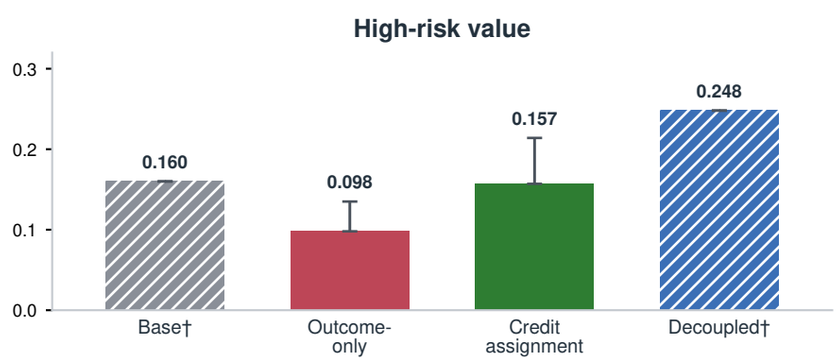
+0.71 reward

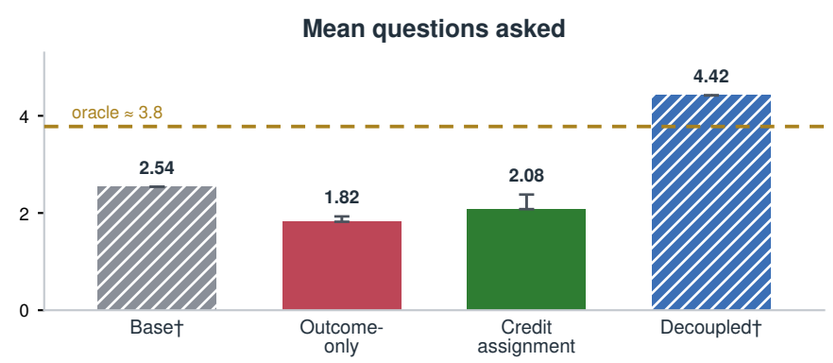
5 non-decisive findings; CT free air and lactate never requested



C Quantitative diagnostics results

held-out diagnostics · same unshaped terminal verifier for every method





High-risk value = mean verifier reward on the rare emergent subset (a behavioural measure, not a direct measure of clinical safety).

† single-seed reference (hatched). Outcome-only / Credit assignment: mean ± SD across 3 training seeds. Dashed line: exact adaptive oracle on the same held-out cases.